

Activity trace — full-development

SHD comparison

ar069 · 18 July 2026 · Draft

Checkpoint CP-001

This mandate-draft checkpoint preserves the visible conversation from the new exp068 request through acceptance and merging of the preceding exp066 night. Its immutable private source is held outside the repository with read-only permissions and SHA-256 prefix `f284e05d1a4a`. The session identifier is `019f76ab-932b-72c2-b163-463ead65d8d5`, the checkpoint identifier is `CP-001`, and the checkpoint time is `2026-07-18 22:47:07.097 UTC`. This checkpoint has no predecessor.

The attachment's private path is replaced with an explicit redaction marker. Injected agent instructions, environment bootstrap material, hidden chain-of-thought, tool-call payloads, credentials, and infrastructure details are excluded. No uncertain sensitive material is published.

Scientific and operational ledger

1. The requested next experiment is a one-seed exploratory comparison, not a confirmatory study. Seed 42 and the complete scientific design are drafted in `ar068` but remain unlocked pending the mandate gate.
2. AUTORESEARCH identified pull request 50 as the unresolved publication gate for exp066 and refused to mix the materially new experiment into that locked night.
3. The scientist authorized merging pull request 50 and opening a new pull request for the new night. GitHub initially refused because the pull request was a draft; it was marked ready and merged with merge commit `5d677ca` at `2026-07-18 22:45:37 UTC`.
4. Preview and focused tests passed for pull request 50. Its repository-wide type check still reports older diagnostics in exp041, exp048, exp054, exp065, and two older tests; this unrelated debt was disclosed before the authorized merge.
5. No new experiment runner, scientific artifact, pod, cost, or official-test observation exists at this checkpoint. The RunPod spend for exp068 is exactly 0 USD and zero pods are active.
6. The accepted branch was synchronized after preserving unrelated generated PDF state in a named local stash. The next unused identifiers are `ar068`, `ar069`, and `exp068`.

Pending work

1. Build and validate the draft mandate and this privacy-sanitized trace.
2. Present the complete exp068 contract and obtain explicit mandate approval.
3. Only after approval, commit the mandate to `main`, create the dedicated night branch, implement and smoke-test exp068, and cross the authorized cloud-compute gate.

4. Maintain timestamped, hash-linked trace checkpoints and pull-request comments at meaningful milestones, then stop at the next publication gate.

Privacy-sanitized visible transcript

User

2026-07-18 22:28:38.331 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Files mentioned by the user:

/goal Design and execute the next exploratory SHD experiment comparing matched ...: [PRIVATE ATTACHMENT PATH REDACTED]

The attached pasted text file(s) contain the user's request. Read and act on that content.

My request for Codex:

Assistant

2026-07-18 22:28:41.960 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'll read the attached `/goal` plus the lab runbook first, then I'll check the current branch and publication gate before changing anything or starting compute.

Assistant

2026-07-18 22:28:44.736 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

`demolab` isn't installed on PATH, so I'm using the repository-prescribed `uvx demolab-cli docs` fallback and will then read the goal in full.

Assistant

2026-07-18 22:28:54.374 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The attached goal is a materially new mandate, and it explicitly preserves PR #50's unresolved publication gate. I'm now following AUTORESEARCH's state inspection; I will not register or run the new experiment if that gate must first be resolved.

Assistant

2026-07-18 22:29:17.139 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR #50 is still open at the human publication gate. AUTORESEARCH requires resolving that gate before registering a materially new mandate, and your goal explicitly says to stop here if so.

Please choose one action for PR #50:

- **Merge** exp066 using a merge commit, accepting its feasibility evidence.
- **Close unmerged**, preserving it as a rejected audit trail.
- **Request changes** only if exp066 itself needs correction.

I have not modified files, posted comments, launched pods, or started exp068.

Assistant

2026-07-18 22:29:20.026 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'm rechecking PR #50 in case its publication state changed externally. If it remains open, the mandate gate is still inaccessible without your review decision.

Assistant

2026-07-18 22:29:27.063 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR #50 remains open and unresolved. I still need your publication decision—**merge**, **close unmerged**, or **request specific exp066 changes**—before AUTORESEARCH permits the new mandate. No files or external state were changed.

Assistant

2026-07-18 22:29:34.873 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Blocked at PR #50's publication gate. Please choose **merge**, **close unmerged**, or **request specific exp066 changes** before the next SHD mandate can be registered.

User

2026-07-18 22:40:52.618 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

continue on pr 50 with a new experiment

Assistant

2026-07-18 22:41:03.527 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

A new exploratory question cannot be added to PR #50 under the locked AUTORESEARCH contract. "Request changes" permits corrections or reruns of exp066; this experiment changes the dataset split, checkpoint selection, training duration, outcome, and interpretation criteria, so it requires a new mandate committed after exp066's publication gate resolves.

Please explicitly choose one:

- **Merge PR #50**, then I'll register and run the new mandate in the proper next-night workflow.
- **Authorize a workflow deviation** allowing the new mandate to continue on PR #50 without first landing its preregistration on `main`.

The first option preserves the requested publication and mandate gates.

User

2026-07-18 22:44:30.939 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

merge or 50 into main, then make a new pr for this autoresearch

Assistant

2026-07-18 22:44:39.977 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'm using the GitHub workflow guidance to verify PR #50's checks and merge readiness, then I'll merge it with a merge commit as authorized. After that I'll return to AUTORESEARCH, reserve the next mandate on `main`, and stop at its approval gate before compute.

Assistant

2026-07-18 22:44:45.211 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR #50 is mergeable at the Git level, but its required lint check is failing while preview and tests pass. I'm following the CI-debugging workflow to determine whether this is a real publication defect or an infrastructure-only failure before merging.

Assistant

2026-07-18 22:45:13.009 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The failure is repo-wide `ty` debt in older experiments (exp041/048/054/065 and two older tests), not exp066 or ar066/ar067. Ruff passes, the focused test workflow passes, and the preview builds. Since you explicitly authorized acceptance, I'm merging PR #50 with a merge commit despite that unrelated blocking check; I won't rewrite history or squash.

Assistant

2026-07-18 22:45:26.968 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The merge operation is still completing; once GitHub confirms it, I'll synchronize `main` and inspect the next unused IDs before drafting the new mandate.

Assistant

2026-07-18 22:45:32.740 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

GitHub refused the merge solely because PR #50 is still marked draft. I'm marking it ready for review as the necessary publication-gate transition, then retrying the authorized merge commit.

Assistant

2026-07-18 22:45:52.250 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR #50 is now merged into `main` as merge commit `5d677ca`. The unrelated repository-wide `ty` failure remains recorded as a known CI issue.

I'm now entering AUTORESEARCH's draft phase: synchronize the accepted evidence, allocate the next unused night/activity/experiment IDs, and prepare—but not commit—the complete new scientific contract for your mandate approval. No compute will start before that gate.

Assistant

2026-07-18 22:46:03.750 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Synchronizing `main` exposed generated PDF changes left by the earlier local build, including untracked exp066 PDFs; Git correctly refused to overwrite them. I'm preserving that entire PDF state in a named stash, then fast-forwarding `main`. I won't discard or commit those unrelated generated changes into the new mandate.

Assistant

2026-07-18 22:46:52.250 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The accepted branch is now synchronized and clean; the next unused identifiers are `ar068` (night mandate), `ar069` (activity trace), and `exp068` (experiment). I'm creating the required immutable private transcript checkpoint now, sanitizing the publishable trace, and drafting those two articles with the complete fixed contract. This trace checkpoint is required by the attached goal; it will not authorize compute or lock the mandate by itself.

Checkpoint CP-002

This mandate-approval checkpoint follows CP-001. Its immutable private source has SHA-256 prefix 5b86314a0d93; the session identifier remains 019f76ab-932b-72c2-b163-463ead65d8d5, and the checkpoint time is 2026-07-18 22:50:59.704 UTC.

Mandate decision

The scientist approved the complete ar068 contract without amendment. This authorizes the preregistration commit on main, followed by creation of the dedicated night branch and

execution within the registered local-smoke, two-pod, three-hour-target, and 10 USD ceilings. No compute had begun and exp068 spend remained 0 USD at approval.

Visible messages added in CP-002

User

2026-07-18 22:50:49.175 UTC

Role: user · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

go

Assistant

2026-07-18 22:50:54.407 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Mandate approved. I'm locking `ar068`, `ar069`, and the reserved `exp068` ID on `main` now, including a hash-linked approval checkpoint. Only after that preregistration commit is pushed will I create the dedicated night branch or run the local smoke.

Checkpoint CP-003

This implementation-and-smoke checkpoint follows the mandate anchor `2ee03ac`. Its immutable private source has SHA-256 prefix `ade35e683d6f`; the session identifier remains `019f76ab-932b-72c2-b163-463ead65d8d5`, and the checkpoint time is `2026-07-18 23:04:58.048 UTC`.

Milestone ledger

1. The approved mandate was committed and pushed to `main` at `2ee03ac`; the night branch was created from that exact anchor.
2. The initial combined cleanup command was rejected locally because it used a force-removal flag. Nothing ran; the build and explicit cleanup then succeeded without discarding user work.
3. The runner initially violated the repository's experiment import gate by importing the engine model module directly. Focused checks caught this before compute. Engine access now routes through the existing tool entrypoint; `tools/snn` remains unchanged.
4. Runner-controlled staging produced a deterministic joint speaker/class split with 7,340 development-training and 816 validation utterances. The official test is unavailable until both checkpoint-freeze flags exist.
5. Experiment-side epoch instrumentation implements the registered selection order: highest validation accuracy, lower validation cross-entropy, then earlier epoch.
6. The local two-epoch smoke passed both matched cells with finite active dynamics and no skipped or non-finite updates. Its accuracy values are plumbing diagnostics only.
7. Targeted Ruff, ty, compilation, and the two-pod dry run pass. No cloud pod has been created; exp068 spend remains 0 USD.

Visible messages added in CP-003

Assistant

2026-07-18 22:51:23.100 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The combined validation command was rejected by the safety wrapper because it contained force-removal syntax for known build byproducts. Nothing ran. I'll rerun the build and cleanup as separate, explicit operations without force flags.

Assistant

2026-07-18 22:52:15.269 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The mandate is now immutably preregistered on `main` at `2ee03ac`, authored solely as Eoin. I'm creating `night/spiking-heidelberg-digits/ar068` from that exact commit. The registered exposure is one local smoke, two concurrent pods maximum, approximately three hours each, and \$10 total; current exp068 spend is \$0 with zero pods active.

Assistant

2026-07-18 22:52:25.660 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The night branch is created from the exact locked commit. I'm mapping the existing SHD loader, trainer outputs, checkpoint behavior, and RunPod helper before implementing; the key requirement is to substitute validation for the trainer's per-epoch "test" split and keep the official test physically unavailable until both selected checkpoints are frozen.

Assistant

2026-07-18 22:55:22.134 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The existing trainer does not preserve every epoch checkpoint, so its default "first strict accuracy improvement" rule cannot satisfy the registered validation tie-break. Without touching `tools/snn`, I'm adding experiment-side checkpoint instrumentation: at each emitted validation epoch, `exp068.py` snapshots the live model only when the registered lexicographic rule improves (accuracy, then loss, then earlier epoch). The runner will replace the CLI's deployment checkpoint with that explicitly selected state before any sealed-test evaluation.

Assistant

2026-07-18 22:58:49.978 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The first focused checks exposed two implementation issues before any compute: unused publication imports, and the repository's experiment gate correctly rejected a direct `models` import. I'm keeping the no-`tools/snn` constraint by routing all engine access through `tools/snn/tool.py`'s existing re-exports, while retaining runner-side split, checkpoint selection, and sealed evaluation. The two-pod dry run itself is correct and created nothing.

Assistant

2026-07-18 22:59:45.213 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The runner now passes targeted Ruff, `ty`, compilation, and the two-pod dry run. I'm starting the registered local 128-train/128-validation, two-epoch plumbing smoke. It will stage the joint speaker/class split, run COBA then PING, enforce the registered checkpoint rule, and restore the host's cached official files afterward. These smoke metrics are not scientific evidence.

Assistant

2026-07-18 23:00:28.893 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The smoke split staged correctly: COBA sees exactly 128 development-training and 128 validation utterances, with the registered 256E/64I configuration and finite initial activity. Epoch 1 is compiling/training now; no official-test data are involved.

Assistant

2026-07-18 23:02:54.398 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

COBA smoke completed cleanly: finite two-epoch training, active E population (~24 Hz), zero I as configured, and the registered selector froze epoch 1 at 11.7% validation accuracy. PING is now training with the same staged indices and its registered 1000 dampening; its initial E/I activity is finite and active. Accuracy remains plumbing-only.

Assistant

2026-07-18 23:04:34.734 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The registered smoke passed both cells with the original locked recipe: COBA selected epoch 1 at 11.72% validation accuracy with 23.93 Hz E activity; PING selected epoch 2 at 10.16% with 2.95 Hz E and 14.56 Hz I activity. All losses and diagnostics were finite, no updates were skipped, and both staged splits used the same index hashes. These numbers are plumbing evidence only. I'm verifying the host's official SHD cache was restored, then checkpointing the implementation and smoke record before any cloud dispatch.

Checkpoint CP-004

This pre-dispatch integrity checkpoint follows implementation commit **ea7b80a**. Its immutable private source has SHA-256 prefix **2ceed244e77d**; the session identifier remains **019f76ab-932b-72c2-b163-463ead65d8d5**, and the checkpoint time is **2026-07-18 23:09:44.045 UTC**.

Integrity audit

The first full staging attempt wrote variable-length event rows in a Python loop and exceeded the local command window. It created no pod, touched no official-test observation,

and produced no scientific result. Bulk HDF5 fancy indexing reduced the same full staging operation to 6.9 seconds. The audited files contain exactly 7,340 development-training and 816 validation utterances, all 20 classes, all 10 official-training speakers, and all 200 speaker-by-class strata in both partitions. The train and validation index hashes match the smoke provenance.

Pull request 51 is open as a draft with the live preview and timestamped smoke comment. RunPod spend remains 0 USD and zero pods are active.

Visible messages added in CP-004

Assistant

2026-07-18 23:05:56.450 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The smoke outcome and engineering corrections are now recorded in the cold-readable Record and CP-003 trace. I'm using the GitHub publish workflow to build, commit, push this exact pre-cloud runner, and open the new draft PR; that gives the live activity log a preview before the authorized compute dispatch.

Assistant

2026-07-18 23:07:43.260 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR #51 is open as a draft with the timestamped smoke checkpoint. Before spending, I'm doing one final non-test-data plumbing audit: generate the full 7,340/816 staged files locally, verify every official-training sample appears exactly once, confirm joint speaker/class coverage, and validate the two pods would see identical index hashes. I will not inspect or evaluate the official test split.

Assistant

2026-07-18 23:08:35.261 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The full staging audit uncovered a real performance issue, not a scientific failure: writing 7,340 variable-length HDF5 rows one by one did not finish within the local command window, leaving only a temporary validation file. The split itself remains valid, but this implementation would waste pod time before training. I'm optimizing the experiment-side HDF5 subset writer in bulk and will rerun the same audit; no pod or official-test access occurred.

Assistant

2026-07-18 23:09:35.726 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The optimized writer completes the full split in 6.9 seconds. Audit results are exact: 7,340 train + 816 validation = 8,156, with all 20 classes, all 10 training speakers, and all 200 speaker×class strata present in both partitions; index hashes match the smoke provenance. I'm committing this staging correction and a trace checkpoint before dispatch.

Checkpoint CP-005

This compute-dispatch checkpoint follows staging commit **96748e5**. Its immutable private source has SHA-256 prefix **db4b6ce7514e**; the session identifier remains **019f76ab-932b-72c2-b163-463ead65d8d5**, and the checkpoint time is **2026-07-18 23:11:55.171 UTC**.

Compute lifecycle

The sealed evaluator passed a two-utterance staged-validation audit. At 2026-07-18 23:11:29 UTC, two RTX 5090 pods were dispatched concurrently from pinned commit **96748e5**, one for each architecture. Their actual rates are 0.99 USD per hour and each has a 10,800-second hard self-removal backstop, limiting actual-rate exposure to 5.94 USD. No pod identifier or sensitive infrastructure detail is published. The official-test seal was closed at dispatch; exp068 spend before dispatch was 0 USD.

Visible messages added in CP-005

Assistant

2026-07-18 23:11:03.633 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The optimized staging checkpoint is pushed at `96748e5`. One high-risk path remains unexercised: the experiment-side sealed evaluator. I'm testing it against two staged validation utterances and the smoke checkpoint—not the official test—so shape, 20-class readout, prediction capture, subgroup logic, and checkpoint loading are proven before cloud spend.

Assistant

2026-07-18 23:11:27.519 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The sealed evaluator passes on staged validation data: it loads the selected 20-class checkpoint, emits per-sample logits and predictions, cross-entropy, accuracy, firing rates, speaker grouping, and runtime without touching official test. The tiny two-sample audit naturally has empty diagnostic groups; the full registered test contains every class and both unseen speakers.

I'm crossing the authorized compute gate now: two RTX 5090 pods, one architecture each, 10,800-second hard self-removal backstops. Maximum ceiling exposure is \$6.54 at the helper's \$1.09/hour cap, below the registered \$10 limit.

Assistant

2026-07-18 23:11:49.877 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both pods are provisioned at the actual rate of \$0.99/hour each, so the three-hour hard-backstop exposure is \$5.94 total. They are still in startup; the official-test seal remains closed. I'm checkpointing this dispatch in ar068/ar069 and PR #51 while the pods run independently under their backstops.

1. Checkpoint CP-006

This full-run and paired-result checkpoint follows CP-005. Its immutable private source has SHA-256 prefix `ab5f70410d5d`; the session identifier remains `019f76ab-932b-72c2-b163-463ead65d8d5`, and the checkpoint time is `2026-07-19 00:29:04.330 UTC`.

Scientific and operational ledger

1. Both fixed 40-epoch cells completed with finite diagnostics and no skipped updates. COBA froze epoch 25; PING froze epoch 40.
2. The official test remained sealed until both checkpoint records existed. The paired evaluation then used all 2,264 official-test utterances exactly once per frozen checkpoint.
3. PING achieved 45.72% accuracy and COBA 35.07%, a registered +10.64 percentage-point difference. PING also improved macro-class accuracy and cross-entropy while reducing excitatory firing. This meets the preregistered promising-signal band but remains one-seed exploratory evidence.
4. Both pods self-removed after their completion markers. Explicit reap and provider-list checks found zero active pods. Measured spend was 2.4414148 USD at the actual 0.99 USD/h rate.
5. The artifact publisher produced provenance, numbers, reproducer, curves, diagnostics, predictions, raw hashes, and matched input/E/I rasters.

Visible messages added in CP-006

Assistant

2026-07-18 23:13:23.128 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The dispatch checkpoint is built, committed, and pushed at ``5548f4b``; the pods remain pinned to pre-dispatch implementation ``96748e5``. They are still cold-starting at roughly one minute, with accrued exposure around \$0.03 total. I'm monitoring the two existing pods only—no duplicate dispatch.

Assistant

2026-07-18 23:14:33.585 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

At about 2.5 minutes both pods report RUNNING, consistent with the normal authenticated image-pull/startup window. No shared artifacts have appeared yet and accrued exposure is about \$0.08 total. The three-hour self-removal backstops remain armed.

Assistant

2026-07-18 23:15:34.302 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both pods are still in provider startup and have not exposed their execution endpoints; no experiment process has begun and no shared output exists. This is a cold-start delay, not a model

failure. At roughly four minutes, exposure is about \$0.13 total, so I'm continuing to monitor the existing pods rather than creating replacements.

Assistant

2026-07-18 23:16:55.302 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Provider state confirms both GPUs are rented but containers have not started (``uptimeSeconds = 0``), so this is provisioning/image startup rather than experiment execution. I'm treating sensitive provider environment fields as private audit material and will not publish them. At roughly seven minutes, exposure is about \$0.23 total; I'll allow the existing authenticated pulls more time under the hard budget.

Assistant

2026-07-18 23:17:46.178 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both containers are now live and each pinned runner is training its assigned architecture. The shared volume contains both live validation metric streams, confirming the two-cell fan-out and common split path are active. The official-test seal is still closed; accrued exposure is about \$0.27 total.

Assistant

2026-07-18 23:18:39.644 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both training processes are active and consuming GPU memory; the first full epoch has not yet completed, which is expected because it includes compilation plus 230 training batches and 26 validation batches. No failure marker or frozen-checkpoint flag exists. I'll report at the first completed validation epoch rather than infer from partial batches.

Assistant

2026-07-18 23:19:16.846 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 1 completed for both cells with finite diagnostics and no skipped updates. Validation accuracy is 20.34% for COBA and 22.79% for PING; PING's validation E rate is 6.01 Hz versus COBA's 23.15 Hz, with active I at 26.79 Hz. These are live validation diagnostics, not sealed-test results. Epoch time is about 111–114 seconds, projecting roughly 75–80 minutes for 40 epochs plus final evaluation—comfortably within the backstops and budget.

Assistant

2026-07-18 23:19:32.965 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

While the fixed cloud runs continue, I'm implementing only the local collection/publication path: cross-cell integrity checks, artifact hashing, figures, and `numbers.json` assembly. This cannot affect the pinned training pods or reveal official-test data early; they remain on `96748e5`.

Assistant

2026-07-18 23:21:37.268 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 2 remains finite with zero skipped/non-finite batches. Validation accuracy is 20.59% COBA and 25.37% PING; the directional gap is validation-only and cannot be interpreted as the registered outcome. Epoch times have settled near 99–102 seconds. The local publisher now passes targeted Ruff, `ty`, compilation, and diff checks, without changing the pinned runs.

Assistant

2026-07-18 23:23:03.530 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 3 completes cleanly: validation reaches 29.53% COBA and 33.46% PING, with PING still using much less E activity (7.18 vs 23.78 Hz). This remains checkpoint-selection data only. The sealed evidence publisher is separately committed at `1fa2c43`; running pods are unaffected and continue from `96748e5`.

Assistant

2026-07-18 23:24:29.648 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 4 remains finite with no skipped updates; both validation accuracies dipped from their epoch-3 best, so the registered selector correctly keeps the earlier checkpoints. I'm switching to a compact 45-second monitor that reports only epoch counts, failure/freeze flags, and active-pod count. This avoids exposing provider details or overinterpreting validation fluctuations.

Assistant

2026-07-18 23:25:52.379 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

At 23:25 UTC both cells remain in epoch 5, with two pods active, no failure marker, and no checkpoint frozen yet. The comparison is progressing normally; the sealed-test barrier remains closed.

Assistant

2026-07-18 23:26:48.627 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells have completed epoch 5. No failure or freeze flag is present, and both pods remain active. I'm letting the fixed protocol continue unchanged; the next detailed metric check will be at a wider milestone unless an integrity flag appears.

Assistant

2026-07-18 23:27:45.697 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

At 23:27 UTC both are still processing epoch 6; the monitor shows no failures, no frozen checkpoint, and two active pods. Current exposure is approximately \$0.51 total, well inside budget.

Assistant

2026-07-18 23:28:39.914 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 6 is complete for both; they are now inside epoch 7. The official-test barrier remains closed and there are still no integrity failures. Estimated exposure is about \$0.56 total.

Assistant

2026-07-18 23:29:34.075 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PING has completed epoch 7 while COBA is finishing its seventh epoch; this small skew is consistent with their observed epoch times. Both pods remain healthy and no sealed-test action has occurred.

Assistant

2026-07-18 23:30:28.211 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are synchronized at 7/40 epochs with two active pods, no failure flags, and no frozen checkpoint. Estimated total exposure is about \$0.61.

Assistant

2026-07-18 23:31:23.129 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PING has completed epoch 8; COBA is finishing epoch 8. No integrity marker has fired, and the official-test seal remains closed.

Assistant

2026-07-18 23:33:12.063 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells were at 8/40 at the last monitor tick, with no failure or freeze flags. I've drafted a neutral, artifact-driven exp068 report whose verdict and every number will come from `numbers.json`; it deliberately makes no pre-result title claim. It cannot build until the sealed artifacts exist and will be validated only afterward.

Assistant

2026-07-18 23:34:07.134 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are at 9/40, still matched in progress with two active pods and no failure/freeze markers. Estimated exposure is about \$0.74 total. The official test remains unavailable.

Assistant

2026-07-18 23:35:13.919 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

At the 10/40 milestone both runs remain finite with zero skipped or non-finite batches. Validation accuracy is 34.56% COBA and 36.64% PING; PING's E rate remains substantially lower (8.78 vs 25.21 Hz) with active I at 39.42 Hz. These diagnostics do not open or predict the registered official-test comparison.

Assistant

2026-07-18 23:36:08.356 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are processing epoch 11; the paired freeze barrier is still closed and no failure flag exists. Total exposure is approximately \$0.80.

Assistant

2026-07-18 23:37:06.646 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells have completed epoch 11. The one malformed local monitor poll was retried immediately and had no effect on either pod. No run failure or protocol anomaly is present.

Assistant

2026-07-18 23:38:01.999 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are at 12/40 with no failures and two active pods. The official-test seal is still closed; estimated exposure is about \$0.88 total.

Assistant

2026-07-18 23:38:56.593 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 13 is in progress for both architectures. No checkpoint has frozen and no failure marker exists. Current exposure remains below \$1.00.

Assistant

2026-07-18 23:39:50.093 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells have completed epoch 13 cleanly. The paired validation phase continues unchanged with the official test sealed.

Assistant

2026-07-18 23:40:44.193 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both are inside epoch 14; no integrity or lifecycle state has changed. Estimated cumulative exposure is about \$0.95.

Assistant

2026-07-18 23:41:39.356 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PING has completed epoch 14; COBA is finishing the same epoch. Both pods remain active with no failure or freeze flag, and sealed evaluation has not begun.

Assistant

2026-07-18 23:42:34.270 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are synchronized at 14/40. Estimated total RunPod exposure is about \$1.00, still far below the \$10 ceiling.

Assistant

2026-07-18 23:43:41.647 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

At 15/40, both remain finite with zero skipped/non-finite batches. The registered selector currently retains COBA epoch 14 at 35.91% validation accuracy and PING epoch 15 at 38.48%. PING's latest E rate is 9.23 Hz versus COBA's 26.15 Hz. These remain validation-only diagnostics; the official-test gate is closed.

Assistant

2026-07-18 23:44:36.356 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are processing epoch 16 with no lifecycle or integrity change. Cumulative exposure is approximately \$1.08.

Assistant

2026-07-18 23:45:33.877 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells have completed epoch 16. No failures, frozen checkpoints, or official-test activity are present.

Assistant

2026-07-18 23:46:28.998 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 17 is in progress for both. Two pods remain active and the run stays within the fixed protocol and budget.

Assistant

2026-07-18 23:47:24.855 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells have completed epoch 17 and are inside epoch 18. No integrity flag has fired; estimated cumulative exposure is about \$1.18.

Assistant

2026-07-18 23:48:19.859 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are at 18/40, with no failure or freeze markers and the official test still sealed.

Assistant

2026-07-18 23:49:16.576 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 19 is in progress on both pods. Exposure is approximately \$1.23 total; the \$10 hard ceiling remains distant.

Assistant

2026-07-18 23:50:13.980 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells have completed epoch 19 without an integrity event. The paired validation phase continues; sealed evaluation has not begun.

Assistant

2026-07-18 23:51:07.951 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both are inside epoch 20 with two active pods, no failure marker, and no frozen checkpoint.

Assistant

2026-07-18 23:52:12.186 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

At 20/40, both remain finite with zero skipped or non-finite batches. The selector currently retains COBA epoch 16 at 38.48% validation accuracy and PING epoch 20 at 42.03%. PING's latest E rate is 9.49 Hz versus COBA's 26.54 Hz. Half the fixed training protocol is complete; the official test remains sealed.

Assistant

2026-07-18 23:53:06.163 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PING has completed epoch 21 while COBA finishes the same epoch. Two pods remain active, with no failure or freeze marker; cumulative exposure is about \$1.37.

Assistant

2026-07-18 23:54:00.360 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are synchronized at 21/40. The validation-only phase remains healthy and sealed evaluation has not begun.

Assistant

2026-07-18 23:54:54.207 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both pods are inside epoch 22 with no integrity state change. Estimated exposure is about \$1.42 total, still well under budget.

Assistant

2026-07-18 23:55:47.887 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells have completed epoch 22. No failure marker, paired freeze, or official-test access is present.

Assistant

2026-07-18 23:56:41.079 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 23 is in progress for both architectures; the fixed protocol continues without intervention.

Assistant

2026-07-18 23:57:35.977 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells have completed epoch 23 and are inside epoch 24. Cumulative exposure is approximately \$1.52; no integrity or lifecycle anomaly is present.

Assistant

2026-07-18 23:58:30.003 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are at 24/40 with two active pods, no failures, and no frozen checkpoints. The official test remains sealed.

Assistant

2026-07-18 23:59:22.991 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 25 is in progress for both. Estimated total exposure is about \$1.57, still far below the hard ceiling.

Assistant

2026-07-19 00:00:27.361 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

At 25/40, diagnostics remain finite with zero skipped/non-finite batches. The selector now retains COBA epoch 25 at 40.81% validation accuracy and PING epoch 23 at 43.38%. Latest E rates are 26.99 Hz COBA and 9.58 Hz PING. These checkpoints remain provisional until all 40 epochs finish.

Assistant

2026-07-19 00:01:31.428 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are processing epoch 26. The date rollover does not change the immutable night ID or registered contract; no integrity state has changed.

Assistant

2026-07-19 00:02:24.862 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells have completed epoch 26, with PING briefly finishing first. No failures or frozen checkpoints exist; exposure is approximately \$1.67 total.

Assistant

2026-07-19 00:03:18.692 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PING has completed epoch 27 while COBA finishes it. The official-test barrier remains closed and both pods remain healthy.

Assistant

2026-07-19 00:04:13.036 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are synchronized at 27/40 with no failure or freeze markers. Estimated cumulative exposure is about \$1.73.

Assistant

2026-07-19 00:05:09.271 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 28 is in progress for both. No intervention, tuning, or sealed-test access has occurred.

Assistant

2026-07-19 00:06:03.599 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells have completed epoch 28. Two pods remain active; no integrity state has changed and the test seal remains closed.

Assistant

2026-07-19 00:06:58.266 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are at 29/40 with no failure/freeze flags. Estimated total exposure is about \$1.83, leaving substantial budget headroom.

Assistant

2026-07-19 00:07:53.448 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 30 is in progress on both pods. The paired freeze barrier remains closed and no anomaly has appeared.

Assistant

2026-07-19 00:08:59.056 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

At 30/40, both runs remain finite with zero skipped/non-finite batches. The selector still retains COBA epoch 25 at 40.81% and PING epoch 23 at 43.38% validation accuracy. Latest E rates are 27.38 Hz COBA and 9.83 Hz PING. Ten fixed epochs remain before checkpoint freeze.

Assistant

2026-07-19 00:09:52.651 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are processing epoch 31. Exposure is approximately \$1.90 total, with no integrity or lifecycle change.

Assistant

2026-07-19 00:10:46.338 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PING has completed epoch 31 while COBA finishes it. The official-test seal remains closed and no failure marker exists.

Assistant

2026-07-19 00:11:42.211 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PING is at 32/40 and COBA at 31/40, a normal sub-epoch skew. Both pods remain healthy; no checkpoint has frozen and sealed evaluation has not begun.

Assistant

2026-07-19 00:12:37.771 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are synchronized at 32/40. Estimated exposure is about \$2.01 total, with no integrity flags.

Assistant

2026-07-19 00:13:38.302 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PING has completed epoch 33 while COBA finishes it. The transient local polling syntax error did not touch either pod; both remain active with the sealed-test barrier closed.

Assistant

2026-07-19 00:14:32.541 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are at 33/40 with no failure or freeze markers. Seven training epochs remain.

Assistant

2026-07-19 00:15:26.586 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 34 is in progress for both architectures. Exposure is approximately \$2.09 total; the run remains well within budget.

Assistant

2026-07-19 00:16:20.399 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells have completed epoch 34 and are inside epoch 35. No integrity state has changed; the official test remains sealed.

Assistant

2026-07-19 00:17:28.634 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

At 35/40, both remain finite with zero skipped/non-finite batches. The provisional selected checkpoints remain COBA epoch 25 at 40.81% and PING epoch 23 at 43.38% validation accuracy. Latest E rates are 27.57 Hz COBA and 9.93 Hz PING. Five fixed epochs remain.

Assistant

2026-07-19 00:18:22.534 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both are processing epoch 36. No failures or frozen checkpoint flags exist; exposure is about \$2.20 total.

Assistant

2026-07-19 00:19:18.962 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells have completed epoch 36 cleanly. Four epochs remain before the paired checkpoint freeze.

Assistant

2026-07-19 00:20:12.985 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 37 is in progress for both. The official-test seal remains closed, and no model-specific action has been taken.

Assistant

2026-07-19 00:21:09.329 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells have completed epoch 37 and are inside epoch 38. Estimated cumulative exposure is about \$2.29.

Assistant

2026-07-19 00:22:03.970 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are at 38/40 with no failures and no frozen checkpoints. Two fixed training epochs remain; the test barrier is still closed.

Assistant

2026-07-19 00:22:57.845 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Epoch 39 is in progress for both architectures. Exposure is approximately \$2.35 total; the comparison remains within all limits.

Assistant

2026-07-19 00:23:51.651 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PING has completed epoch 39; COBA is finishing it. No integrity marker has fired and sealed evaluation remains blocked.

Assistant

2026-07-19 00:24:47.595 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are synchronized at 39/40. The final fixed training epoch is now in progress; neither checkpoint has frozen yet.

Assistant

2026-07-19 00:25:42.130 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PING has completed epoch 40 and is finalizing its validation artifacts; COBA is still finishing epoch 40. The paired freeze barrier correctly remains closed until both checkpoint records exist.

Assistant

2026-07-19 00:26:45.613 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'm at the paired completion barrier now. I'm continuing the existing monitor, then I'll collect both sealed official-test results together, reap and verify zero pods, publish the evidence bundle, and finish PR #51 at the human-review gate.

Assistant

2026-07-19 00:27:03.033 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both pods have self-terminated, so the one-pod-per-cell ceiling is satisfied and there is no active cloud exposure. I'm collecting the two result bundles without opening either result in isolation; the next scientific read will be paired.

Assistant

2026-07-19 00:27:57.953 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The paired result is decisive at this exploratory scale: PING reached 45.72% official-test accuracy versus COBA's 35.07%, a +10.64 percentage-point advantage. PING also improved macro-class accuracy and cross-entropy; the sealed test contained all 2,264 official examples. Both result files were present before disclosure, and a post-run reap check confirms zero pods.

Assistant

2026-07-19 00:29:04.330 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The artifact publisher now has a complete provenance bundle: locked split hashes, frozen checkpoint hashes, all validation/activity curves, official-test diagnostics, predictions, matched input/E/I rasters, reproducer, and raw-file hashes. Measured exposure was \$2.4414148 total, well below the \$10 ceiling. I'm updating the timestamped research record and validating the cold-readable report now.